

Herramientas matemáticas para el Análisis Multivariante con R

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#Vectores

Para generar un vector:

```
x<-c(2,3,4)
y<-3:5
z<-seq(from=1, to=6,by=2)
q<-rep(1,3)
x
```

```
## [1] 2 3 4
```

```
y
```

```
## [1] 3 4 5
```

```
z
```

```
## [1] 1 3 5
```

```
q
```

```
## [1] 1 1 1
```

#Acceso a los elementos de un vector

```
x[1]
```

```
## [1] 2
```

```
x[1:2]
```

```
## [1] 2 3
```

```
x[-1]
```

```
## [1] 3 4
```

#Operaciones con vectores

```
#Suma  
x+y
```

```
## [1] 5 7 9
```

```
#Diferencia  
x-y
```

```
## [1] -1 -1 -1
```

```
#Multiplicación  
x*y
```

```
## [1] 6 12 20
```

```
#División  
x/y
```

```
## [1] 0.6666667 0.7500000 0.8000000
```

```
#Resto de una división  
x%%y
```

```
## [1] 0 0 0
```

```
#Vector traspuesto  
t(x)
```

```
##      [,1] [,2] [,3]  
## [1,]    2    3    4
```

```
#Producto interno o escalar  
x%%*%y
```

```
##      [,1]  
## [1,]   38
```

```
t(x)%*%y
```

```
##      [,1]  
## [1,]   38
```

```
t(y)%*%x
```

```
##      [,1]  
## [1,]   38
```

```
#Longitud de un vector  
length(x)
```

```
## [1] 3
```

```
#Multiplicación de un escalar por un vector  
2*x
```

```
## [1] 4 6 8
```

```
##Suma de elementos de un vector mediante producto escalar
```

```
t(x)%*%rep(1,length(x))
```

```
##      [,1]  
## [1,]    9
```

```
##Media de un vector mediante producto escalar
```

```
n<-length(x)  
t(x)%*%rep(1,length(x))/n
```

```
##      [,1]  
## [1,]    3
```

```
##Suma de cuadrados de los elementos de un vector
```

```
t(x)%*%x
```

```
##      [,1]  
## [1,]   29
```

```
##Varianza
```

```
(t(x)%*%x/n)-(t(x)%*%rep(1,length(x))/n)^2
```

```
##      [,1]  
## [1,] 0.6666667
```

```
##Matrices
```

```
Para generar una matriz:
```

```
A<-matrix(data=1:6 ,nrow=2, ncol=3)  
A
```

```
##      [,1] [,2] [,3]  
## [1,]    1    3    5  
## [2,]    2    4    6
```

```
B<-matrix(data=12:17 ,nrow=2, ncol=3)
B
```

```
##      [,1] [,2] [,3]
## [1,]  12  14  16
## [2,]  13  15  17
```

Para generar un data frame a partir de vectores:

```
peso<-c(45,56,78)
altura<-c(156,167,170)
datos<-data.frame(peso,altura)
datos
```

```
##   peso altura
## 1   45    156
## 2   56    167
## 3   78    170
```

```
#Nombre de columnas
colnames(datos)
```

```
## [1] "peso"  "altura"
```

```
#Nombre de filas
rownames(datos)
```

```
## [1] "1" "2" "3"
```

Acceso a elementos de una matriz

```
#Fila
A[2,]
```

```
## [1] 2 4 6
```

```
#Columna
A[,2]
```

```
## [1] 3 4
```

```
#Columna sin simplificar
A[,2,drop=FALSE]
```

```
##      [,1]
## [1,]    3
## [2,]    4
```

```
#Elemento
```

```
A[2,2]
```

```
## [1] 4
```

```
#Subconjunto de elementos
```

```
A[1:2,2:3]
```

```
##      [,1] [,2]
```

```
## [1,]    3    5
```

```
## [2,]    4    6
```

```
A[1:2,c(1,3)]
```

```
##      [,1] [,2]
```

```
## [1,]    1    5
```

```
## [2,]    2    6
```

```
A[, -1]
```

```
##      [,1] [,2]
```

```
## [1,]    3    5
```

```
## [2,]    4    6
```

```
#Operaciones con matrices
```

```
#Suma
```

```
A+B
```

```
##      [,1] [,2] [,3]
```

```
## [1,]   13   17   21
```

```
## [2,]   15   19   23
```

```
#Diferencia
```

```
A-B
```

```
##      [,1] [,2] [,3]
```

```
## [1,]  -11  -11  -11
```

```
## [2,]  -11  -11  -11
```

```
#Multiplicación elemento a elemento
```

```
A*B
```

```
##      [,1] [,2] [,3]
```

```
## [1,]   12   42   80
```

```
## [2,]   26   60  102
```

```
#Multiplicación matricial (columnas de A=filas de B)  
A*%*t(B)
```

```
##      [,1] [,2]  
## [1,]  134  143  
## [2,]  176  188
```

```
#División  
A/B
```

```
##      [,1]      [,2]      [,3]  
## [1,] 0.08333333 0.2142857 0.3125000  
## [2,] 0.15384615 0.2666667 0.3529412
```

```
#Resto de una división  
A%/%B
```

```
##      [,1] [,2] [,3]  
## [1,]    0    0    0  
## [2,]    0    0    0
```

```
#Dimensión de una matriz  
dim(A)
```

```
## [1] 2 3
```

```
##Primer número es número de filas y segundo número de columnas  
nrow(A)
```

```
## [1] 2
```

```
ncol(A)
```

```
## [1] 3
```

```
#Combinar filas y columnas
```

```
#Filas  
rbind(x,y)
```

```
##      [,1] [,2] [,3]  
## x      2    3    4  
## y      3    4    5
```

```
#columnas  
cbind(x,y)
```

```
##      x y
## [1,] 2 3
## [2,] 3 4
## [3,] 4 5
```

```
##Traspuesta
```

```
t(A)
```

```
##      [,1] [,2]
## [1,]    1    2
## [2,]    3    4
## [3,]    5    6
```

```
##Suma y medias de filas y columnas
```

```
#Suma por filas
rowSums(A)
```

```
## [1]  9 12
```

```
#Suma por columnas
colSums(A)
```

```
## [1]  3  7 11
```

```
#Media por filas
rowMeans(A)
```

```
## [1]  3  4
```

```
#Media por columnas
colMeans(A)
```

```
## [1] 1.5 3.5 5.5
```

```
##Determinante
```

```
C<-matrix(data=c(1,5,6,3,8,6,2,3,1),nrow=3,ncol=3)
det(C)
```

```
## [1] -7
```

```
##Traza
```

```
traza<-function(X)
{
  traza<-0
  for (i in 1:dim(X)[1])
    traza<-traza+X[i,i]
  return(traza)
}
traza(C)
```

```
## [1] 10
```

```
#Extraer elementos de la diagonal de una matriz cuadrada
```

```
diag(C)
```

```
## [1] 1 8 1
```

```
##Traza
```

```
sum(diag(C))
```

```
## [1] 10
```

```
##Inversa
```

```
solve(C)
```

```
##      [,1]      [,2] [,3]
## [1,]  1.428571 -1.285714  1
## [2,] -1.857143  1.571429 -1
## [3,]  2.571429 -1.714286  1
```

```
##Vector de medias
```

```
n<-nrow(A)
vector_medias<-(t(A)%*%rep(1, n))/n
vector_medias
```

```
##      [,1]
## [1,]  1.5
## [2,]  3.5
## [3,]  5.5
```

```
##Matriz centrada
```

```
J<-matrix(rep(1,n^2), nrow=n, ncol=n)
J
```

```
##      [,1] [,2]
## [1,]    1    1
## [2,]    1    1
```

```
A_cent<-A-(J%*%A)/n
A_cent
```

```
##      [,1] [,2] [,3]
## [1,] -0.5 -0.5 -0.5
## [2,]  0.5  0.5  0.5
```

```
##Matriz de dobles productos
```



```
P<-t(A)%*%A
P
```

```
##      [,1] [,2] [,3]
## [1,]    5   11   17
## [2,]   11   25   39
## [3,]   17   39   61
```

```
#Con las variables centradas
C<-t(A_cent)%*%A_cent
C
```

```
##      [,1] [,2] [,3]
## [1,]  0.5  0.5  0.5
## [2,]  0.5  0.5  0.5
## [3,]  0.5  0.5  0.5
```

```
CC<-t(A)%*%A-n*vector_medias)%*%t(vector_medias)
CC
```

```
##      [,1] [,2] [,3]
## [1,]  0.5  0.5  0.5
## [2,]  0.5  0.5  0.5
## [3,]  0.5  0.5  0.5
```

```
##Matriz de varianzas-covarianzas
```

```
S<-CC/(n-1)
S
```

```
##      [,1] [,2] [,3]
## [1,]  0.5  0.5  0.5
## [2,]  0.5  0.5  0.5
## [3,]  0.5  0.5  0.5
```

```
##Matriz de correlaciones
```

```
Ds<-diag(diag(S)^(1/2))
Ds
```

```
##      [,1]      [,2]      [,3]
## [1,] 0.7071068 0.0000000 0.0000000
## [2,] 0.0000000 0.7071068 0.0000000
## [3,] 0.0000000 0.0000000 0.7071068
```

```
R<-solve(Ds)%*%S)%*%solve(Ds)
R
```

```
##      [,1] [,2] [,3]
## [1,]    1    1    1
## [2,]    1    1    1
## [3,]    1    1    1
```

##Descomposición espectral (matrices cuadradas)

```
eigen(S)
```

```
## eigen() decomposition
## $values
## [1] 1.500000e+00 4.440892e-16 0.000000e+00
##
## $vectors
##      [,1]      [,2]      [,3]
## [1,] -0.5773503  0.8164966  0.0000000
## [2,] -0.5773503 -0.4082483 -0.7071068
## [3,] -0.5773503 -0.4082483  0.7071068
```

##Descomposición en valores singulares (matrices rectangulares)

```
svd(A)
```

```
## $d
## [1] 9.5255181 0.5143006
##
## $u
##      [,1]      [,2]
## [1,] -0.6196295 -0.7848945
## [2,] -0.7848945  0.6196295
##
## $v
##      [,1]      [,2]
## [1,] -0.2298477  0.8834610
## [2,] -0.5247448  0.2407825
## [3,] -0.8196419 -0.4018960
```

```
X3 <- matrix(c(1, 1,
               1, 1,
               1,-2), ncol = 2, byrow = TRUE)
X3
```

```
##      [,1] [,2]
## [1,]    1    1
## [2,]    1    1
## [3,]    1   -2
```

```
n3 <- nrow(X3)
p3 <- ncol(X3)
```

```
XtX <- t(X3) %*% X3
XtX
```

```
##      [,1] [,2]
## [1,]    3    0
## [2,]    0    6
```

```
eig <- eigen(XtX)
lambda <- eig$values
V <- eig$vectors
lambda
```

```
## [1] 6 3
```

```
V
```

```
##      [,1] [,2]
## [1,]    0  -1
## [2,]    1   0
```

```
alpha <- sqrt(lambda)
alpha
```

```
## [1] 2.449490 1.732051
```

```
Sigma <- diag(alpha)           # (p x p)
Sigma_inv <- diag(1/alpha)     # (p x p), aquí alpha > 0

U <- X3 %*% V %*% Sigma_inv    # (n x p)
U
```

```
##      [,1] [,2]
## [1,] 0.4082483 -0.5773503
## [2,] 0.4082483 -0.5773503
## [3,] -0.8164966 -0.5773503
```

```
t(U) %*% U
```

```
##      [,1] [,2]
## [1,]    1    0
## [2,]    0    1
```

```
t(V) %*% V
```

```
##      [,1] [,2]
## [1,]    1    0
## [2,]    0    1
```

```
X3_rec <- U %*% Sigma %*% t(V)
X3_rec
```

```
##      [,1] [,2]
## [1,]    1    1
## [2,]    1    1
## [3,]    1   -2
```

```
all.equal(X3, X3_rec)
```

```
## [1] TRUE
```